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22nd May 2021

# Probabilistic Machine Learning
# Session 1

# Probability Theory

## Single and Joint Probabilities

Probability theory typically describes random variables, $P(X)$, and the probability that these random variables take on actual values.

$$P(D = 6) = 1/6$$

$$P(D_1 = 6, D_2 = 6) = 1/36$$

## Independent Probabilities

When events are **independent** their **joint** probability is given by the product of their individual probabilities.

$$P(D_1 = 6, D_2 = 6) = 1/36$$

$$P(D_1 = 6, D_2 = 6) = P(D_1 = 6) \times P(D_2 = 6)$$

## Conditional Probabilities

**Conditional** probabilities describe the probability of any event **given** the occurrence of another event.

$$P(D = 6 | D \text{ is even}) = 1/3$$

$$P(D_1 = 6 | D_1 + D_2 > 8) = \ ?$$

## The Rules of Probabilities

There are just two fundamental rules of probability theory.

**Sum rule** $$P(X) = \sum_Y P(X, Y)$$

**Product rule** $$P(X, Y) = P(Y|X) \times P(X)$$

Plus the observation that probabilities must sum to 1.

**Frequentist probability** is the interpretation of probability that defines an event's probability as the limit of its relative frequency in a large number of trials.

**Bayesian probability** is the interpretation of the probability that holds that it can be defined as the degree to which a person (or community) believes that a proposition is true.

Rev. Thomas Bayes (1702 - 1761)

- Framework for plausible reasoning developed over the nineteenth century
- Developed by three physicists:
  - Harold Jeffreys (1891 – 1989)
  - Richard Cox (1898 - 1991)
  - Edwin Jaynes (1922 – 1998)
- Growth of computational power since 2000 has seen this become one of the dominant model for much of machine learning.
- Recent development of probabilistic programming make using Bayesian reasoning ever more accessible.

Cox's axioms describe desirable properties of a framework for reasoning about uncertainty

1. Degrees of plausibility are represented by real numbers.
2. Qualitative correspondence with common sense:
   - Increasing the plausibility of a statement, decreases the plausibility of its negative.
3. Consistency:
   - If a conclusion can be reasoned out in more than one way, then every possible way must lead to the same result.
   - It must be possible to incorporate all the evidence relevant to a question into the reasoning process.
   - All equivalent states of knowledge are represented by equivalent plausibility assignments.

Bayes Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

## Bayes Theorem

Posterior Likelihood Prior

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

Evidence

## Example

- A clinical test, designed to diagnose a specific illness, comes out positive for a certain patient.
- We are told that
  - The test is 79 percent reliable (that is, it misses 21 percent of actual cases)
  - On average, this illness affects 1 percent of the population in the same age group as the patient
  - The test has a false positive rate of 10 percent.
- Taking this into account and assuming you know nothing about the patient's symptoms or signs, what is the probability that this patient actually has the illness?

## Example

$$P(\mathrm{D}) = 0.01$$

$$P(\mathrm{T}|\mathrm{D}) = 0.79$$

$$P(\mathrm{T}|\neg\mathrm{D}) = 0.1$$

## Example

$$\begin{aligned} P(\mathrm{D}|\mathrm{T}) &= \frac{P(\mathrm{T}|\mathrm{D})P(\mathrm{D})}{P(\mathrm{T})} \\ &= \frac{P(\mathrm{T}|\mathrm{D})P(\mathrm{D})}{P(\mathrm{T}|\mathrm{D})P(\mathrm{D}) + P(\mathrm{T}|\neg\mathrm{D})P(\neg\mathrm{D})} \\ &= \frac{0.79 \times 0.01}{0.79 \times 0.01 + 0.1 \times 0.99} \\ &\approx 0.074 \end{aligned}$$

## Next Time

Bayesian Inference and
Markov Chain Monte Carlo